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Completed	Friday, 16 October 2026, 11:09 AM
Duration	4 mins 22 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The lower middle element is chosen as the middle element
c. The search stops
d. The higher middle element is chosen as the middle element ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. Number values should only be present

Number values should only be present

c. The list must be sorted
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search continues until the end of the list
c. The search starts over from the beginning
d. The search stops immediately ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is very small
c. When the list contains duplicate elements
d. When the list is sorted ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O👎
c. O(log n)
d. O(1) ✓

Question **6**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 10 ✓
c. 1000
d. 500

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Quick search
c. Merge search
d. Bubble search

Question **8**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear

b. Binary ✓

c. Hash

d. Both (1) & (3)

Question **9**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓

b. The last element is returned

c. The first element is returned

d. An error is raised

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.

b. Linear search requires the list to be sorted.

c. Linear search is more efficient than binary search on large lists.

d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓

b. When the list is sorted

c. When the list contains duplicate elements

d. When the search algorithm finishes

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The list is sorted
- c. The search stops
- d. The search continues from the beginning

Question **13**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Hash search
- b. Both (1) & (3)
- c. Linear search ✓
- d. Binary search

Question **14**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The list is sorted
- b. The search continues in the left sublist
- c. The search continues in the right sublist
- d. The search ends successfully ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search has a lower time complexity on large, sorted lists ✓
- b. Binary search does not require dividing the list
- c. Binary search works on unsorted lists
- d. Binary search can find multiple instances of the target element